MG: Non-Determinism + Probability.

Side quests: - Why Monad?
- Algebra

Monads: Model computations

state of model: values → Object A

next computations: TA

want: $\eta : A \Rightarrow TA$

Given $f : A \to TB$ } Moggi says:
want $f^* : TA \to TB$ } This is Monad

Non-determinism monad: Pfin

Now: Giry "+" Pfin?

Quest: How does probability interact with non-determinism?

Simplifications: Giry and Pfin on Meas/QBS
Dfin and Pfin on Meas/QBS
Dfin and Pfin on Set

New Goal: This is impossible

Algebra

Σ-Algebras (New)

Σ set of operations with arity $\{+^{(2)}, 0^{(0)}\}$

Terms with variables in X

- variable is a term
- apply operation to terms to form a new term

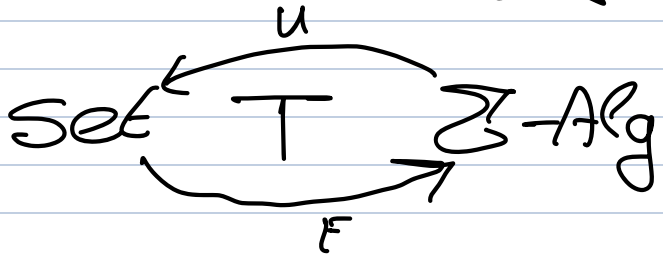
$a+b$ $\quad$ $(a+b)+0$

E set of equations : $\{ a+0 = a$
$(a+b)+c = a+(b+c) \}$

Models for Σ, E are sets with right structure

e.g: $\mathbb{N}, +, 0$, $\mathbb{N}, \cdot, 1$

Models form a category of Σ-Alg.



$U \circ F$ is monad, the Σ, E is called the presentation of this monad.

Presentation Pfin: $\Sigma = \{+^{(2)}, 0^{(0)}\}$

$E = \{ a+0 = a$, $0+a = 0$
associativity
$a+b = b+a$
$a+a = a \}$

Presentation of Dfn $\Sigma = \{+^p \mid p \in (0,1)\}$

$E = \{ a +^p a = a$

$a +^p B = B +^{1-p} a$

$a +^p (B +^r C) =$

$(a + \frac{p}{p+(1-p)r} B) +^{p+(1-p)r} C\}$

$(a + B) +^{\frac{1}{2}} (c+d) = (a +^{\frac{1}{2}} c) +$

$(a +^{\frac{1}{2}} d) +$

$(B +^{\frac{1}{2}} c) +$

$(B +^{\frac{1}{2}} d)$

$(a+B) +^{\frac{1}{2}} (a+B) = a+B$

$(a +^{\frac{1}{2}} a) + (a +^{\frac{1}{2}} B) + (B +^{\frac{1}{2}} a) + (B +^{\frac{1}{2}} B)$

$a + (a +^{\frac{1}{2}} B) + B$

Solutions : • instead of sets, take convex sets.

↳ literature to comp.

• valuations instead prob.

↳ Varacca & Winskel